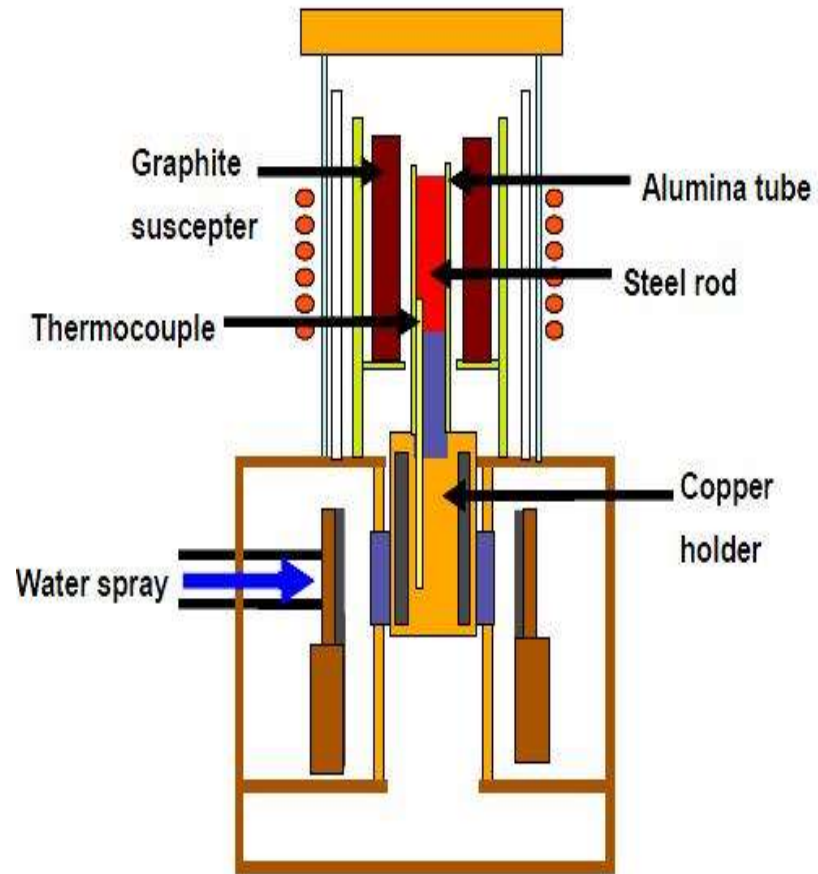


Solidification Structures

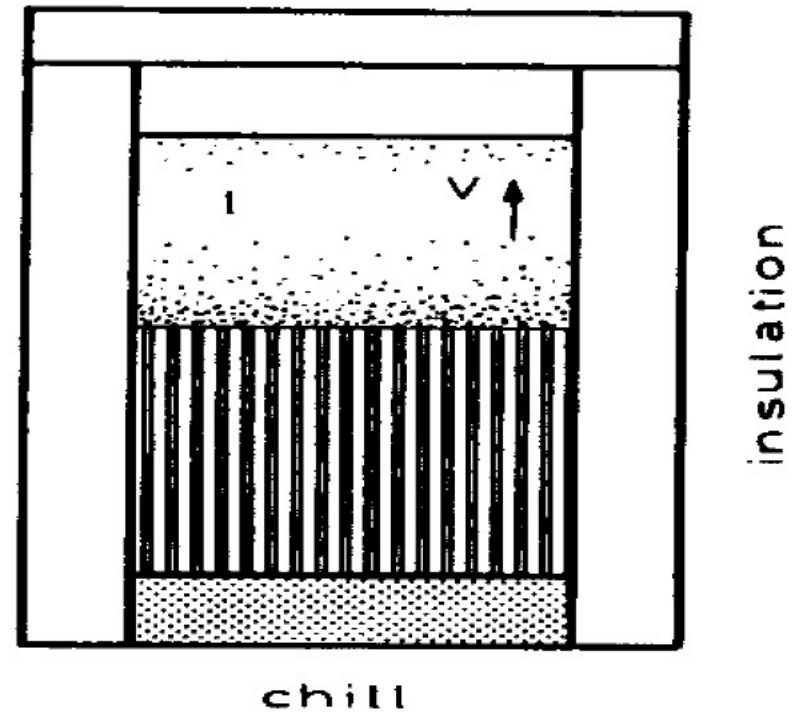
Macro Structure

Micro Structure

Heat Extraction

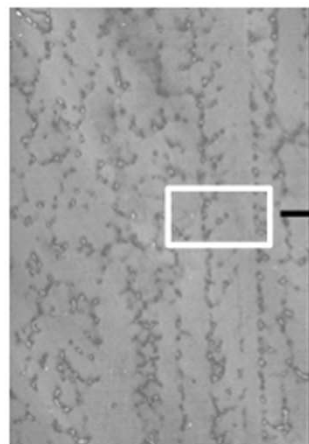
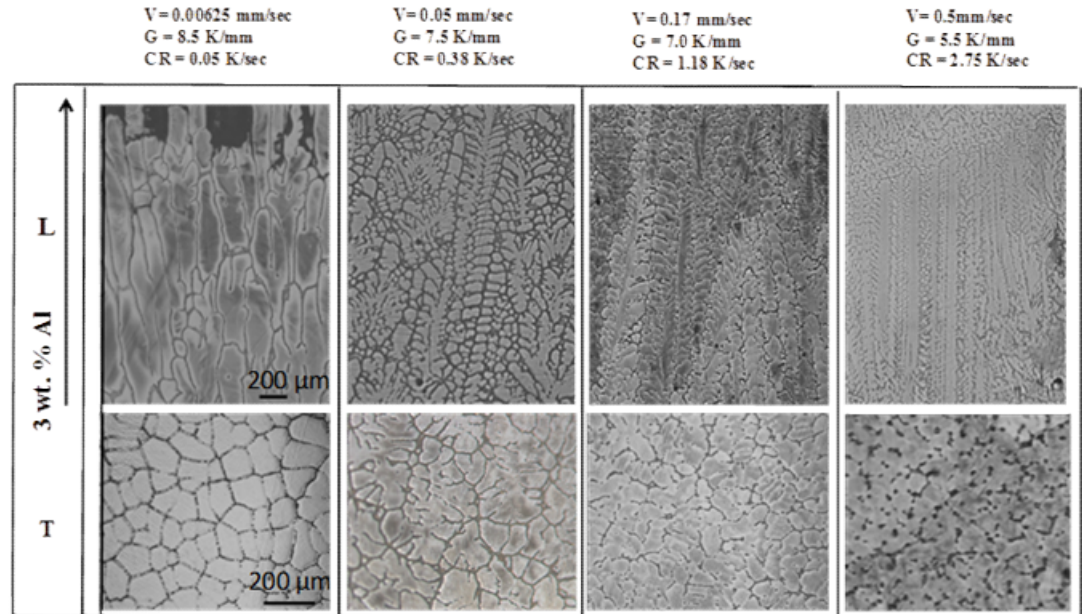
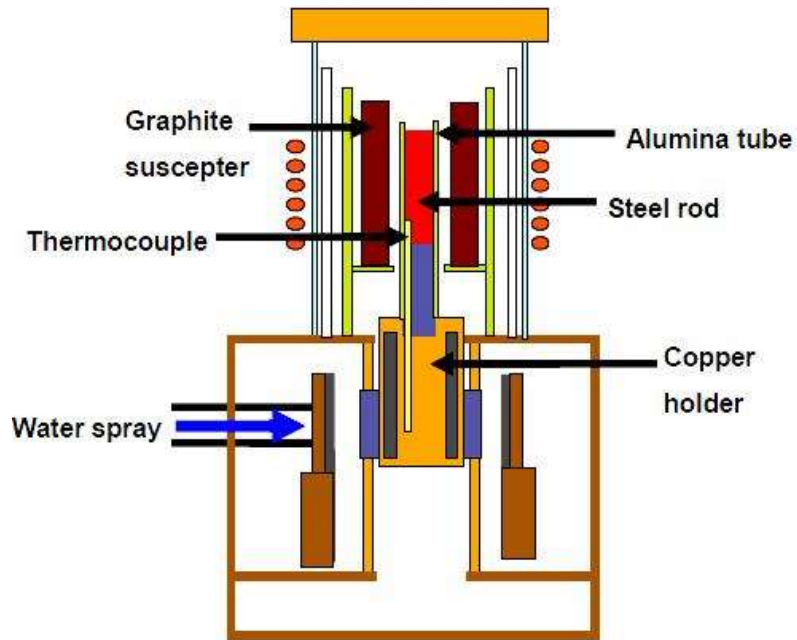


Unidirectional solidification

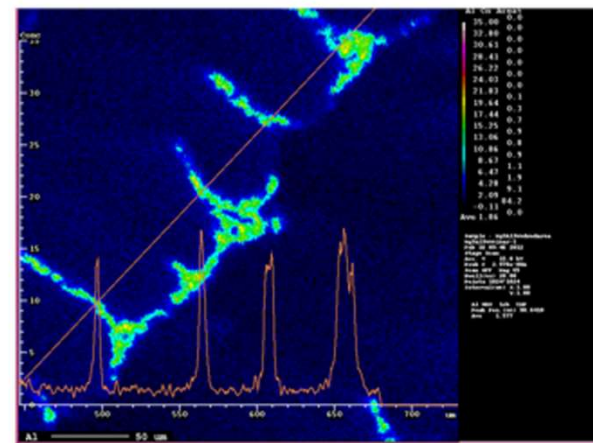


Chill casting

Observation of different growth morphologies: Mg- 3 wt.% Al



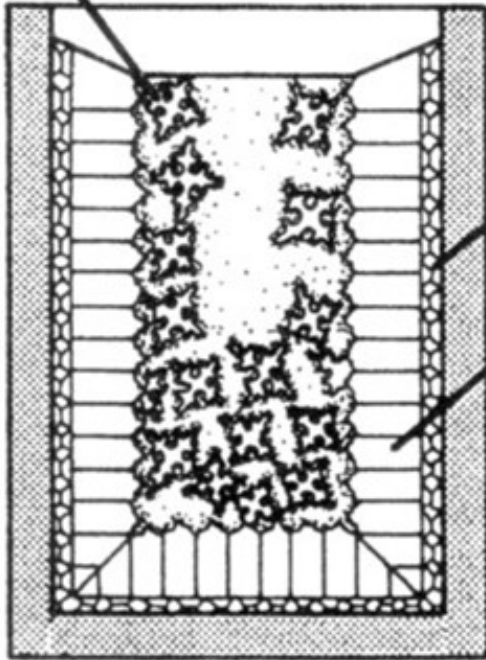
(a)



(b)

Solidification Macrostructures

Equiaxed grain



- Macrostructures in a casting are
Chill grain,
Columnar grain,
and Equiaxed grain.
- Most of them are composed of dendrites.

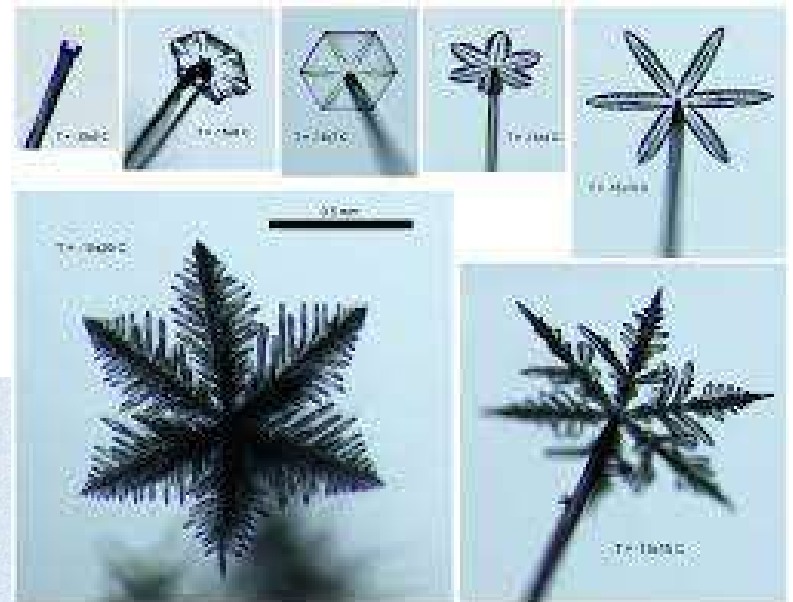
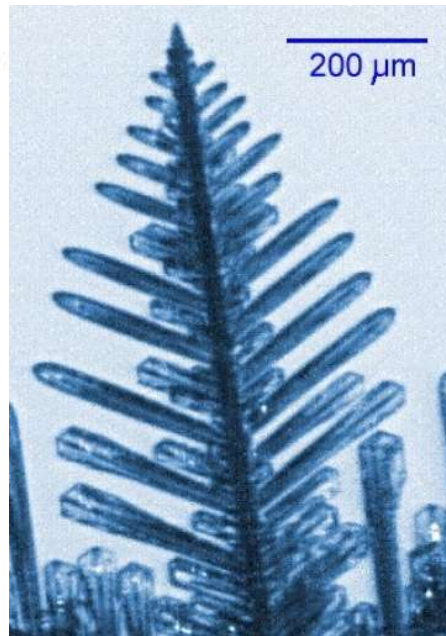


Microstructures

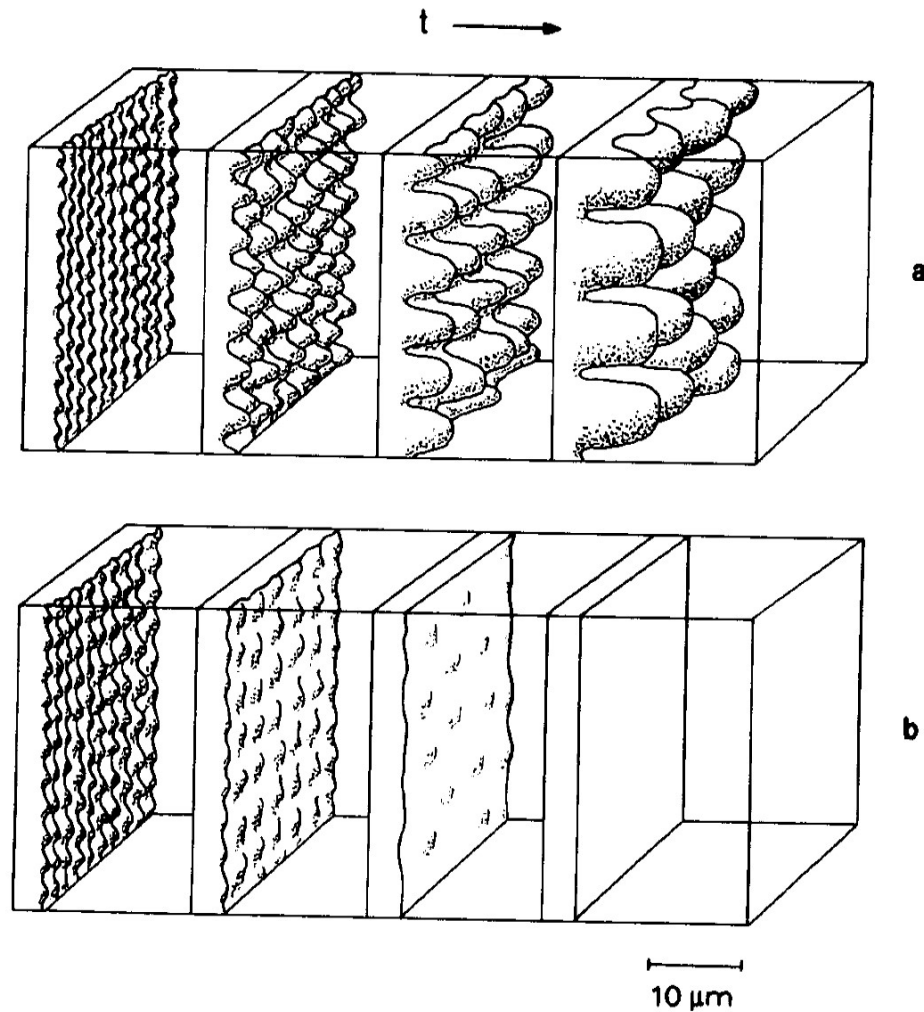
Most of microstructures are of arrayed or single dendrites, and eutectics.



Dendrite

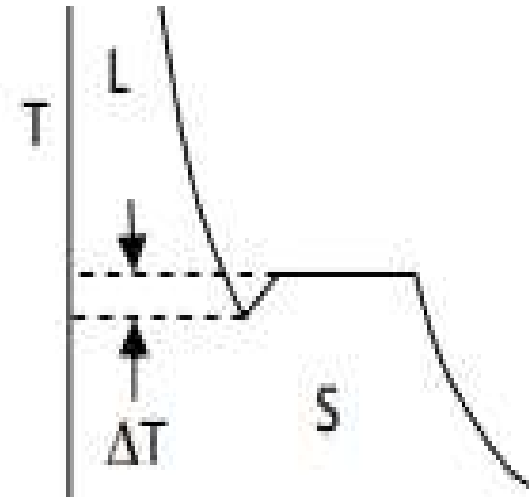
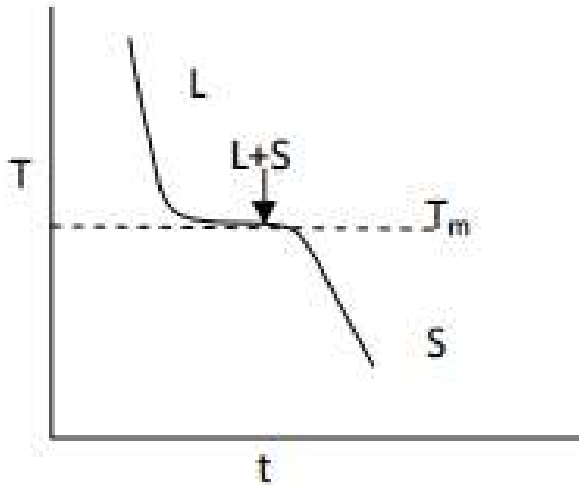


Solidification analysis by instability

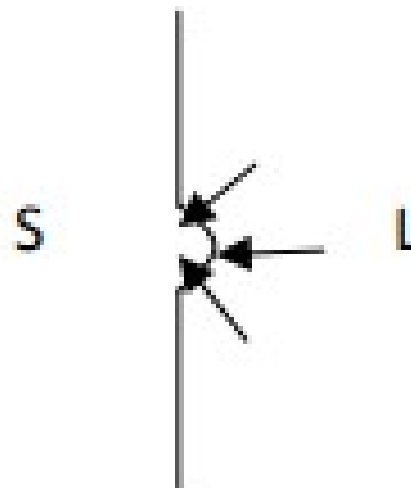


Solidification of pure metals

Cooling curves



Stability analysis



Alloys

Common nomenclatures

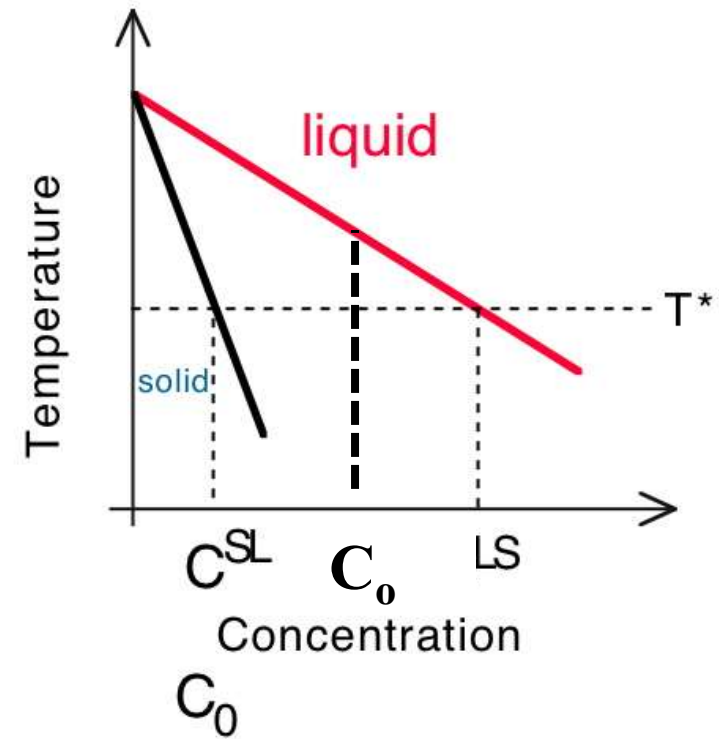
* k: partition coefficient = $(C_s/C_l)_{T,P}$

* m: liquidus slope = (dT/dC_l)

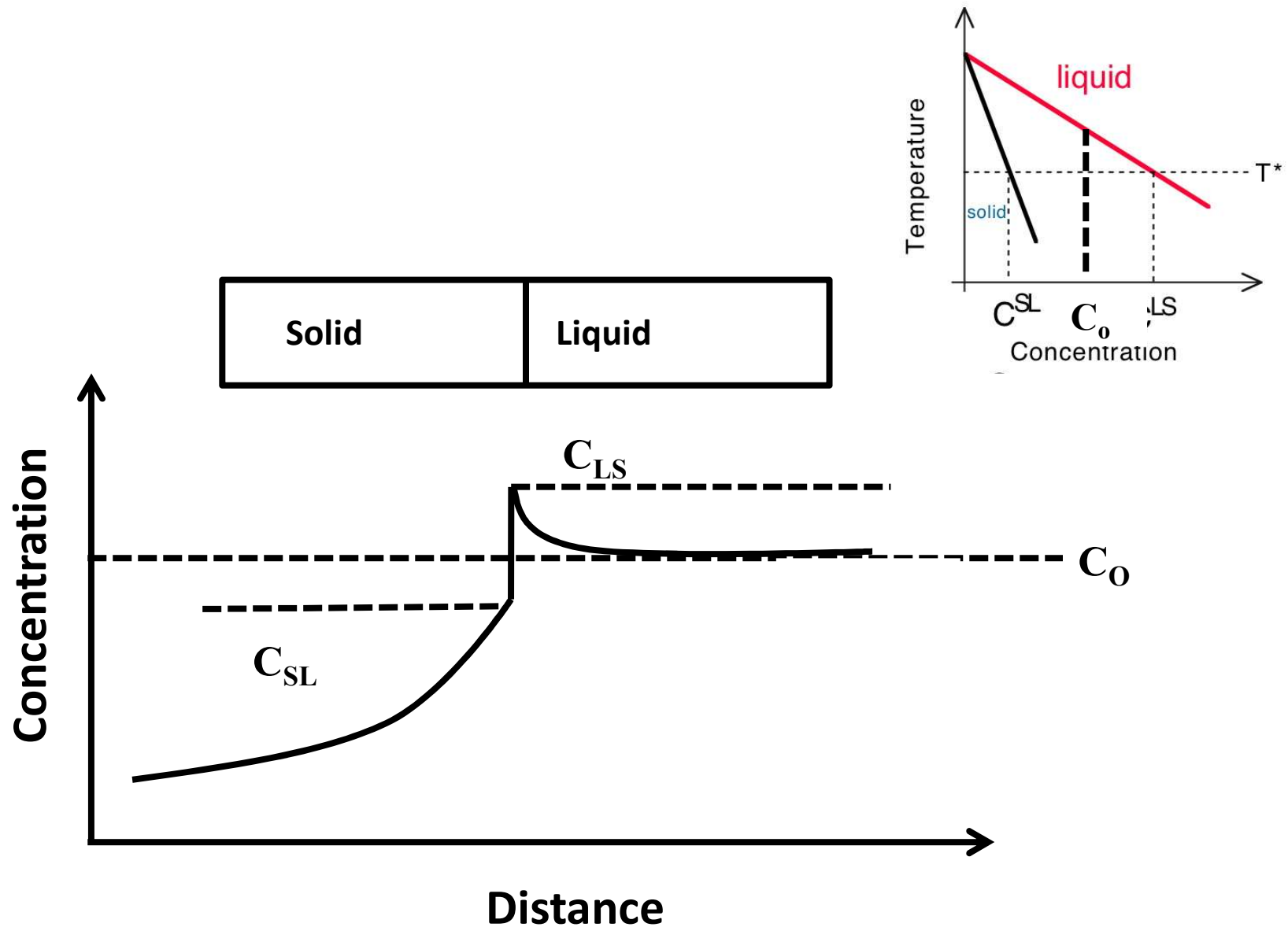
* Thermal gradient: $G_l = (dT_l/dx)$ or $G_s = (dT_s/dx)$

* Imposed thermal gradient: G

* Solidification velocity: V



Solute pile up: planar solidification



Historical topics

Morphology → size → structure

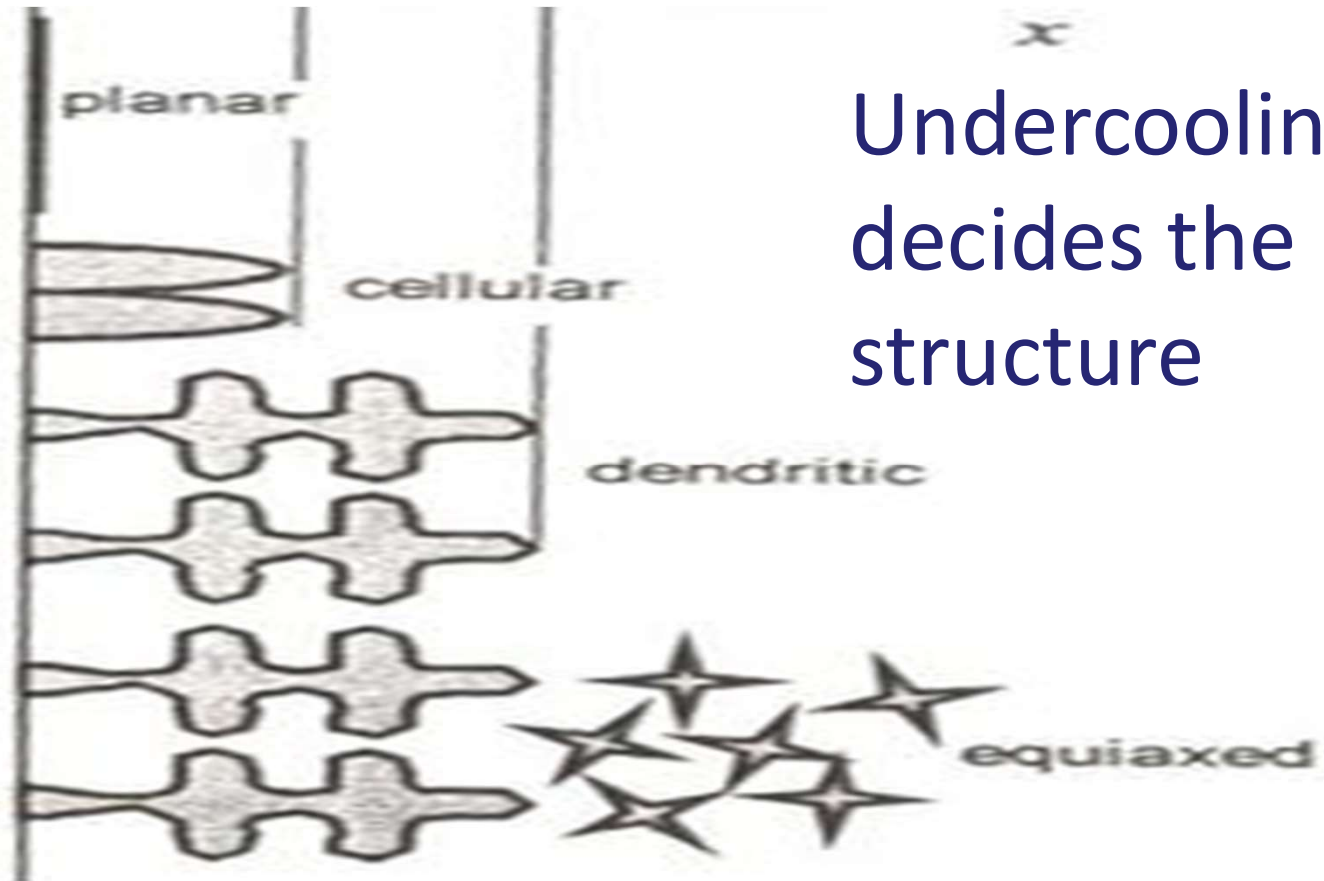
Year	Theory	Experiments	Dendrite growth
1947			Ivantsov's solution
1950	Maximun undercooling		
1953	Constitutional undercooling theory		
1958			
1960			
1963	MS theory		
1966	Secondary arm spacing		
1974			
1976		SNC dendrite	
1978	Marginal stability criterion		
1980			Kurz-Fisher model
1985			
1986	Phase-field model		KGTmodel

Textbook

- B. Chalmers, Principles of Solidification (1964)
- M. C. Flemings, Solidification Processing (1974)
- W. Kurz & D. J. Fisher, Fundamentals of Solidification (1986)

Constitutional undercooling theory

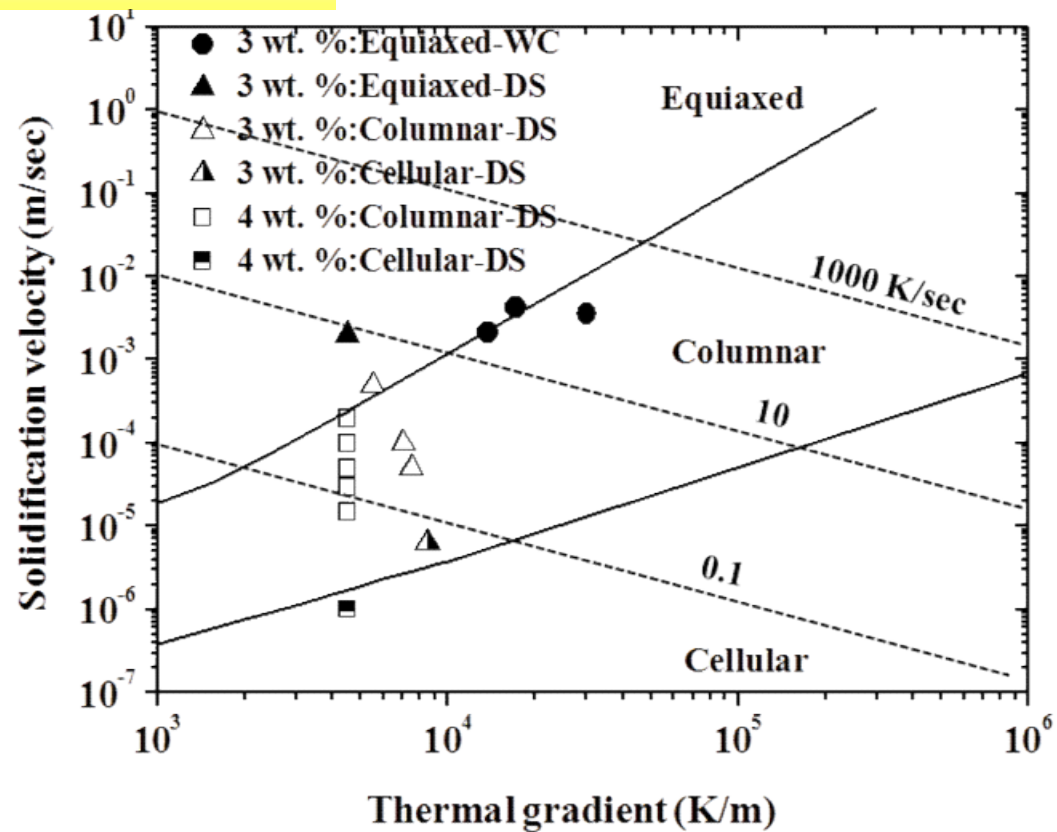
Planar front



Undercooling
decides the
structure

Morphology and solidification conditions

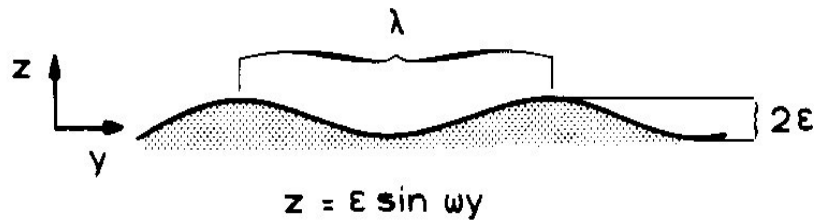
Solidification structures are explained using the parameters of G and V .



Perturbation theory by Mullin & Sekerka



R.F. Sekerka

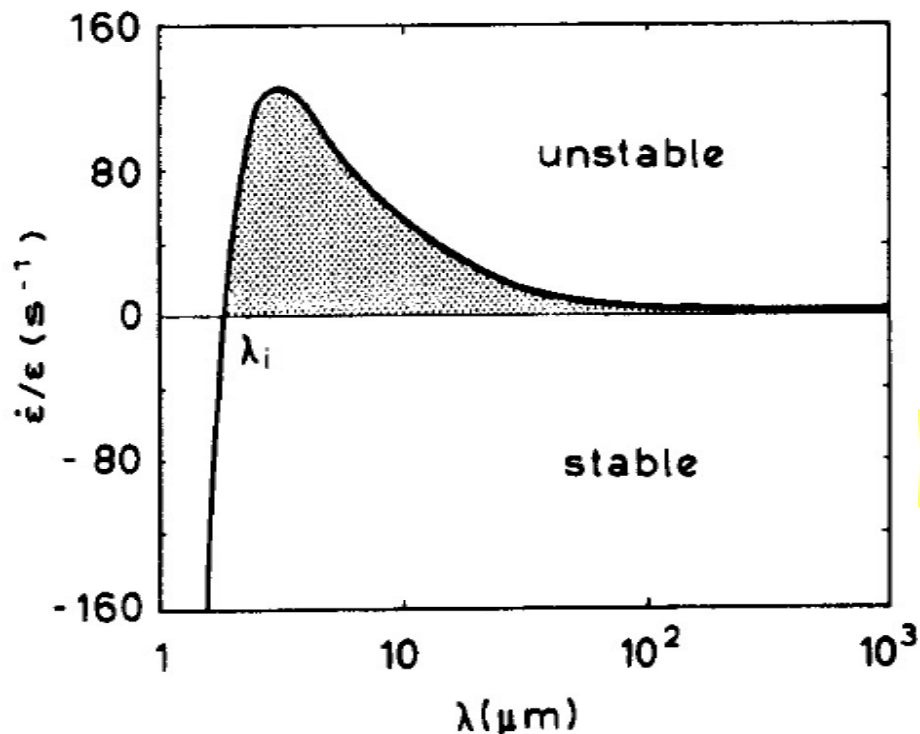


Stable condition of planar interface growth is deduced using linear stability analysis

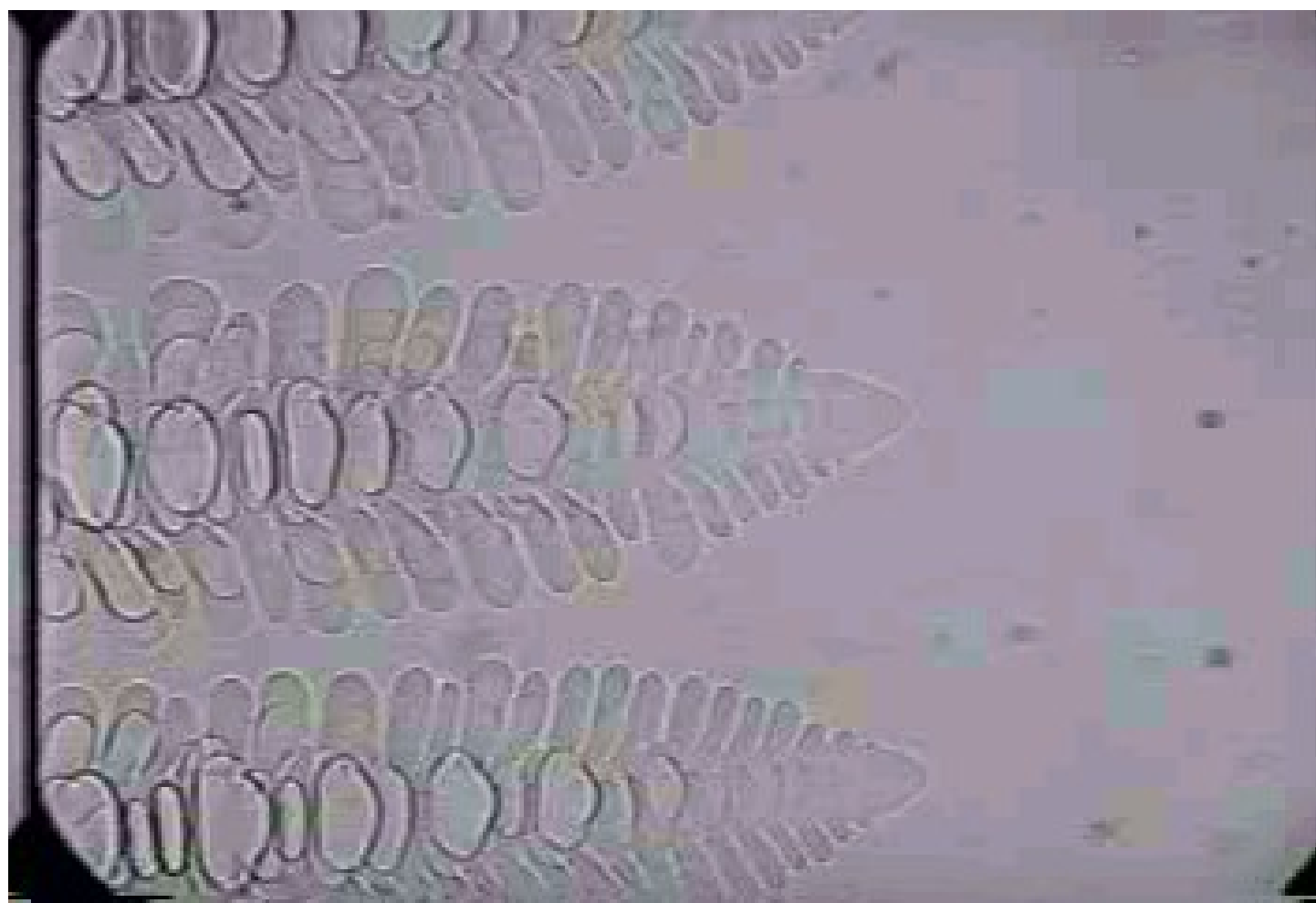
When the amplitude velocity of perturbation waves is negative for all wave length, the planar interface is stable.

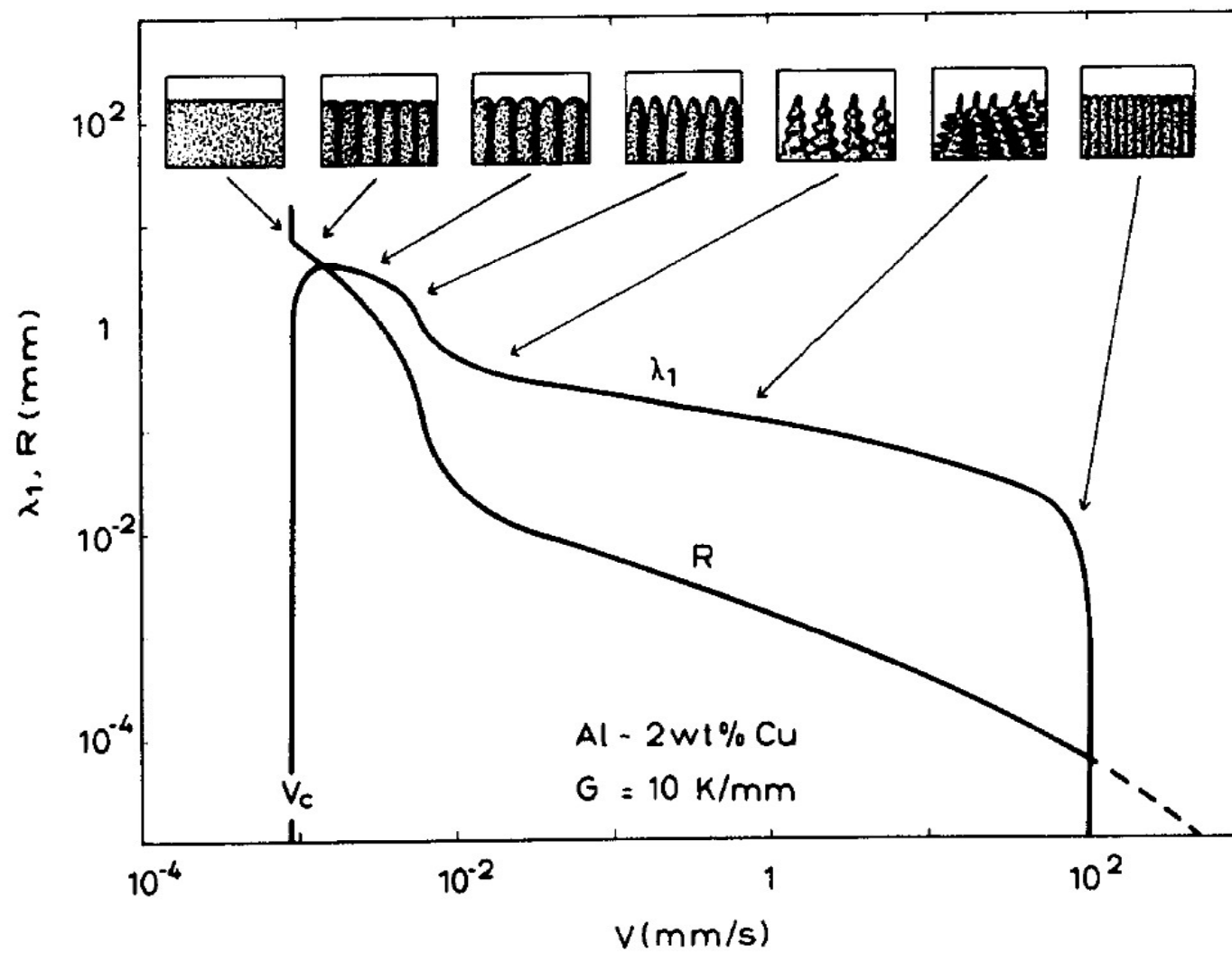


To Marginal stability criterion



Mullins & Sekerka (1963, 1964)



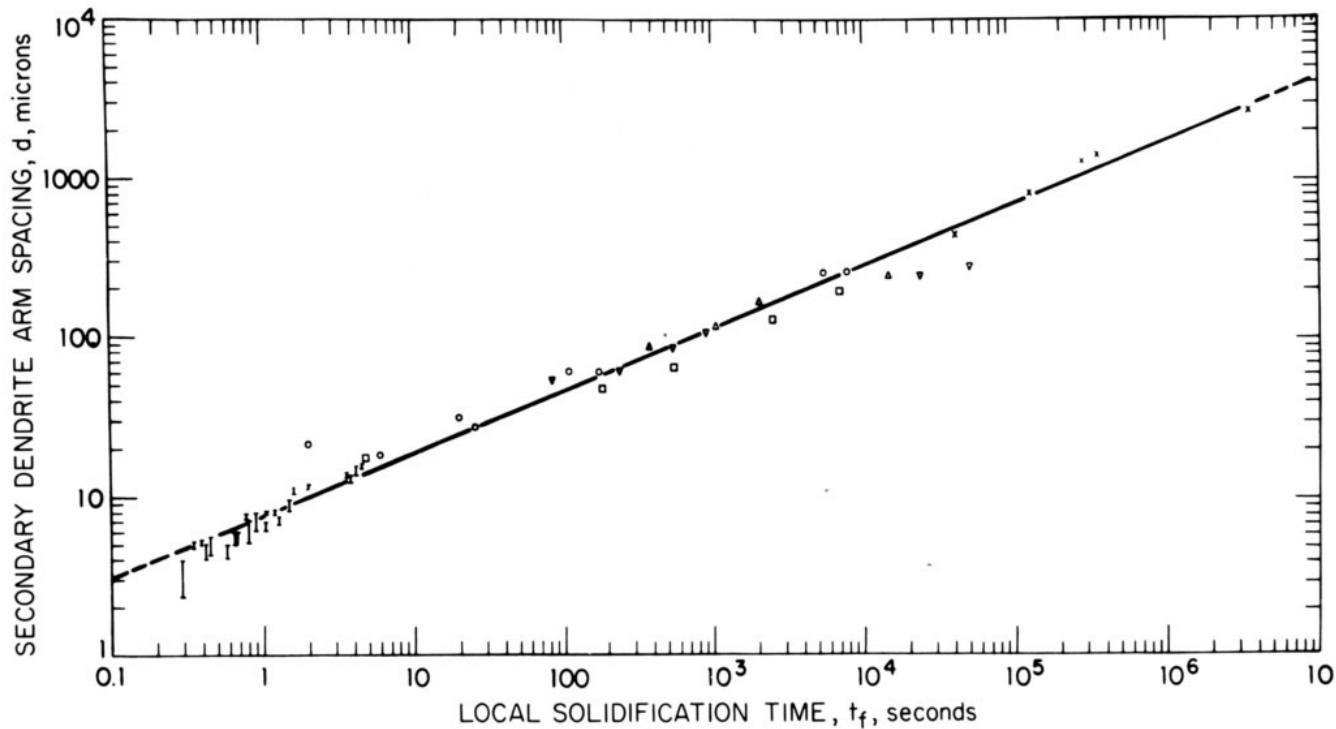


Dendrite arm spacing as a domain unit

Measurement of secondary arm spacing



M.C. Flemings

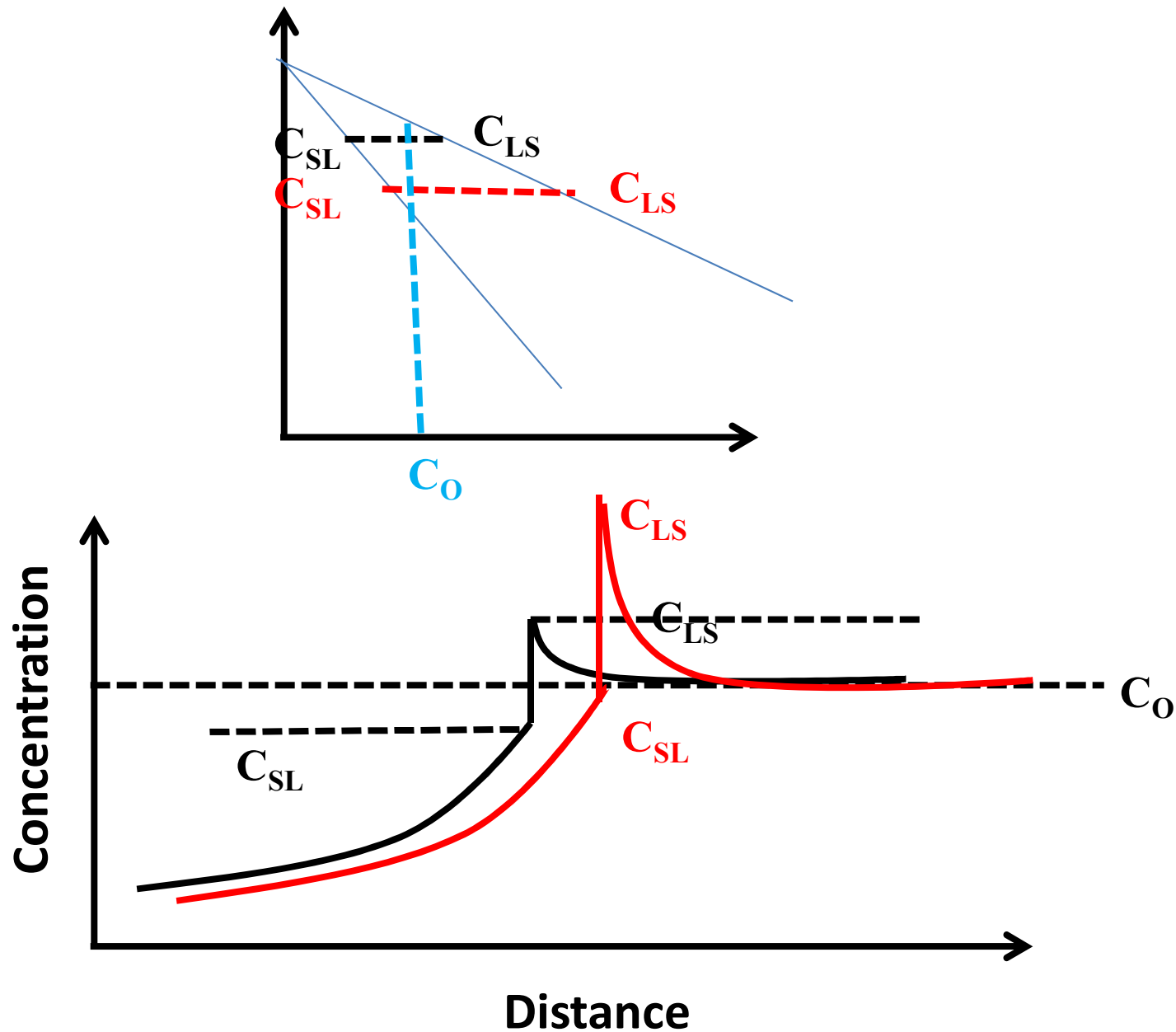


Structure size is determined by solidification conditions.

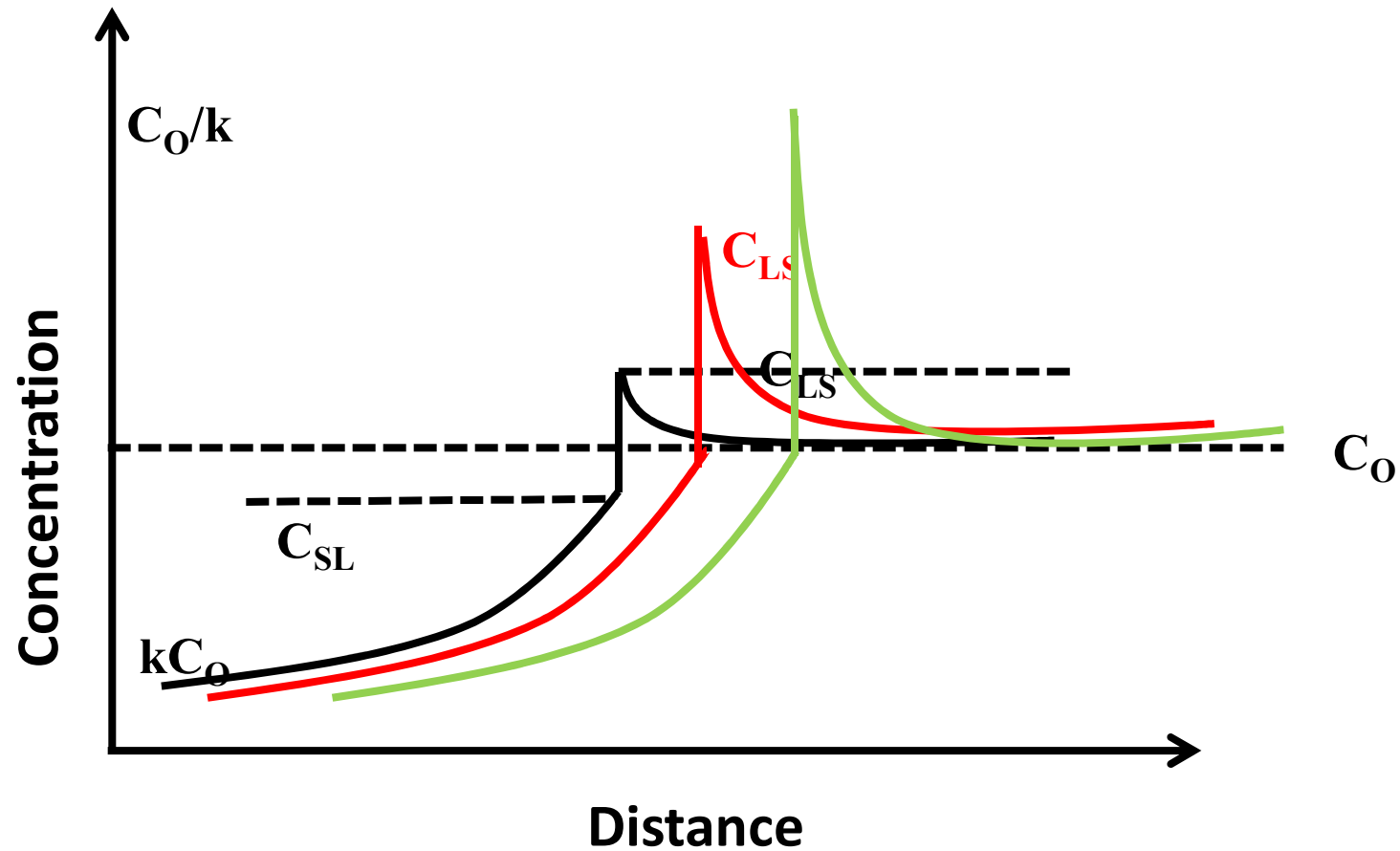
Before the work, dendrite arm spacing is believed to be dependent on the alloy and the casting process.

Bower, Brody & Flemings (1966)
Kattamis, Coughlin & Flemings (1967)

Solute pile up: planar solidification

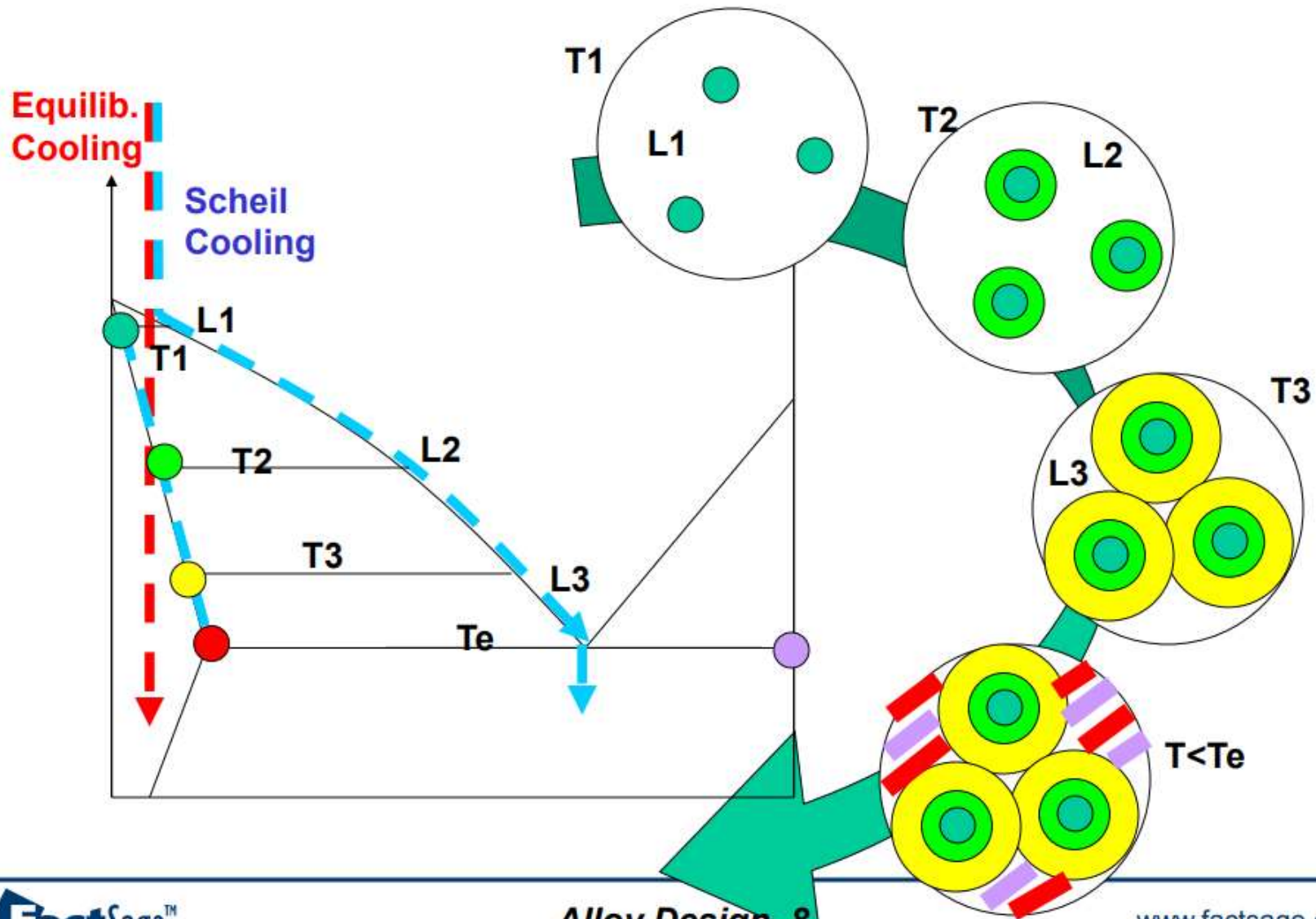


Solute pile up: planar solidification

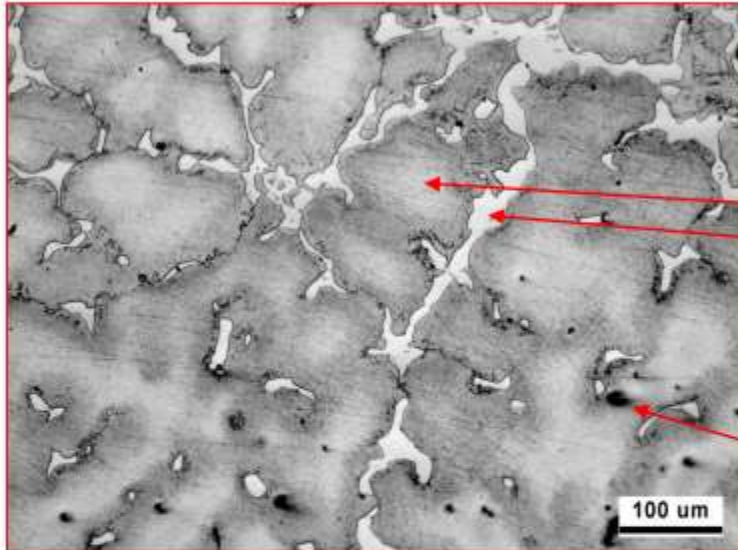


— Steady state

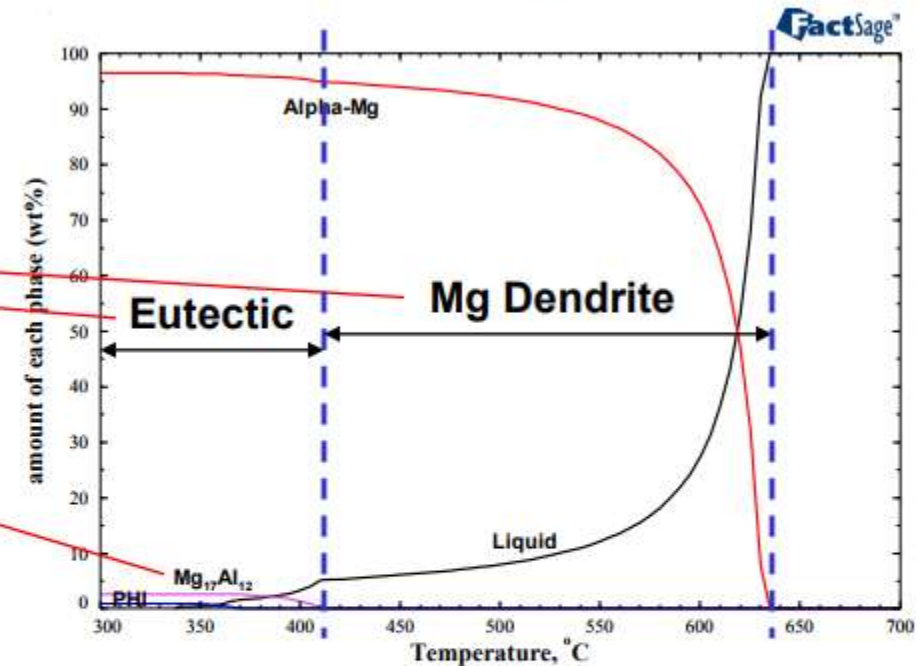
Alloy design (II): Estimation of as-cast microstructure



Solidification path calculation: AZ31



Microstructure of as-cast AZ31



Scheil Cooling calculation

Scheil cooling calculations tell us the solidification path

- Primary phase, precipitation, and eutectic reactions.
- Dendrite boundary (eutectics, segregation)

-> Estimation of as-cast microstructure